113705015 鄭大勝 HW0415

C11501Q8

8. Factor: the type of fertilizer

Levels: 3 types of fertilizer being used

Treatment: 3 types of fertilizer 粉

JOYKO® 36 Lines, 6 mm

C11S01 Q9

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9. Factor: Daily dose of vitamin C
Levels: the 3 doses: 200, 500, 1000 mg
Treatments: the 3 doses: 200, 500, 1000 mg

C11S01 Q14

14. I would not assume that it's normal because it has a fixed number of trials (n=100) and you are counting the "success" which in this case it's detected. So, it's actually a binomial distribution.

C11S02 Q8

8. a.

Source	SS	df	MS	F
Treatment	641.88	2	320.94	5.15
Error	498.67	8	62.33	
Total	1140.55	10		

$\bar{x}_A = 76$

$\bar{x}_B = 66.33$

$\bar{x}_C = 87$

$\bar{G} = 76.36$

Total SS $= \sum_{i=1}^{n} (x_i - \bar{G})^2 = 1140.55$

SST $= \sum_i ((\bar{x}_i - \bar{G})^2 \times k) = 641.88$

SSE $= 1140.55 - 641.88 = 498.67$

b. (1) $H_0: \mu_1 = \mu_2 = \mu_3$

H_A: at least 1 treatment mean is different

(2) $\alpha = 0.05$

(3) $F = \frac{MST}{MSE} \sim F_{2,8;0.05}$

(4) $F^* = 5.15 > F_{2,8;0.05} = 4.46$

(5) R.R: $\{F \mid F > F_{2,8;0.05}\}$

(6) $\therefore$ Reject H_0 $\Rightarrow$ yes, there is a difference in mean based on the data.

C11S02 Q10

10. a. $\alpha = 0.05$

P-value $= 0.0251 < 0.05$ $\therefore$ Reject H_0

$\alpha = 0.01$

P-value $= 0.0251 > 0.05$

$\therefore$ Non-reject H_0

b. 99% CI:

$(\bar{x}_A - \bar{x}_B) \pm t_{\alpha/2} \cdot \sqrt{MSE \cdot \left(\frac{1}{n_A} + \frac{1}{n_B}\right)} = 9.833 \pm 3.250 \times \sqrt{14.941 \times \left(\frac{1}{4} + \frac{1}{3}\right)}$

$\Rightarrow (-3.76, 15.43)$

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c. 99% CI :

$$\bar{x}_A \pm t_{\alpha/2} \cdot \sqrt{\frac{MSE}{n_1}} = 60.5 \pm 3.250 \times \sqrt{\frac{14.941}{4}}$$

$$\Rightarrow (54.22, 66.78)$$

d. We can't be perfectly confident because the sample size is small. However, assembly times are continuous measurements and aren't clustered near a boundary, so it is a reasonable approximation, and ANOVA is robust enough to handle minor deviations

C11S02Q14

14. a.

Source	SS	df	MS	F
Treatment	0.9224	3	0.3075	1.49
Error	6.8104	33	0.2064	
Total	7.7328	36		

b. ① $H_0: \mu_1 = \mu_2 = \mu_3$

H_α: at least 1 treatment mean is different

② $\alpha = 0.05$ and $\alpha = 0.01$

③ $F = \frac{MST}{MSE} \sim F_{k-1, n-k}$

④ $F^* = 1.49$

⑤ $\alpha = 0.05$

$R.R: \{F \mid F > \overset{2.89}{F_{3,33;0.05}}\}$

$\alpha = 0.01$

$R.R: \{F \mid F > \overset{4.44}{F_{3,33;0.01}}\}$

⑥ ∴ Non-reject H_0 for $\alpha = 0.05$ and $\alpha = 0.01$

c. 95% CI :

$$(\bar{x}_1 - \bar{x}_4) \pm t_{\alpha/2, df} \sqrt{MSE\left(\frac{1}{n_1} + \frac{1}{n_4}\right)} = -0.251 \pm 2.035 \times \sqrt{0.2064 \times \left(\frac{1}{14} + \frac{1}{11}\right)}$$

$$\Rightarrow (-0.623, 0.121)$$

d. 95% CI :

$$(\bar{x}_2 - \bar{x}_3) \pm t_{\alpha/2; df} \times \sqrt{MSE\left(\frac{1}{n_2} + \frac{1}{n_3}\right)} = 0.055 \pm 2.035 \times \sqrt{0.2064\left(\frac{1}{8} + \frac{1}{4}\right)}$$

$$\Rightarrow (-0.511, 0.621)$$

e. Light Tuna in water & white tuna in water
Light Tuna in water & white tuna in oil
White tuna in water & light tuna in oil
white tuna in water & light tuna in oil

C11S03 Q16

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16. $W = q_\alpha(k, df) \times \frac{s}{\sqrt{n}} = q_{0.01}(4, 16) \times \sqrt{\frac{91.25}{5}} = 14.07$

$|\bar{x}_1 - \bar{x}_2| = |262.2 - 234.8| = 27.4 > 14.07 \Rightarrow$ differ

$|\bar{x}_1 - \bar{x}_3| = |262.2 - 251.6| = 10.6 < 14.07 \Rightarrow$ not differ

$|\bar{x}_1 - \bar{x}_4| = |262.2 - 268.6| = 6.4 < 14.07 \Rightarrow$ not differ

$|\bar{x}_2 - \bar{x}_3| = |234.8 - 251.6| = 16.8 > 14.07 \Rightarrow$ differ

$|\bar{x}_2 - \bar{x}_4| = |234.8 - 268.6| = 33.8 > 14.07 \Rightarrow$ differ

$|\bar{x}_3 - \bar{x}_4| = |251.6 - 268.6| = 17 > 14.07 \Rightarrow$ differ

17. a.

Source	SS	df	MS	F
Treatment	74,806.67	2	37403.33	4.47
Error	225920	27	8367.41	
total	300,700.67	29		

(1) $H_0: \mu_1 = \mu_2 = \mu_3$

H_A: at least 1 treatment mean is different

(2) $\alpha = 0.05$

(3) $F = \frac{MST}{MSE} \sim F_{2,27;0.05}$

(4) $F^* = 4.47 > F_{2,27;0.05} = 3.35$

(5) R.R: $\{F > F_{2,27;0.05}\}$

(6) $\therefore$ Reject H_0

b. 95% CI:

$$(\bar{x}_L - \bar{x}_P) \pm t_{\alpha/2, df} \times \sqrt{MSE \times \left(\frac{1}{n_L} + \frac{1}{n_P}\right)} = -45 \pm 2.052 \times \sqrt{8367.41 \times \left(\frac{1}{10}\right)}$$

$\Rightarrow (-128.95, 38.95)$

c. $W = q_\alpha(k, df) \times \frac{s}{\sqrt{n}} = 3.51 \times \sqrt{\frac{8367.41}{10}}$

$= 101.54$

$|\bar{x}_L - \bar{x}_P| = |608 - 653| = 45 < 101.54 \Rightarrow$ not differ

$|\bar{x}_L - \bar{x}_S| = |608 - 532| = 76 < 101.54 \Rightarrow$ not differ

$|\bar{x}_P - \bar{x}_S| = |653 - 532| = 121 > 101.54 \Rightarrow$ differ

C11S04 Q24

24. a. Treatments : 4 stores, Blocks : 8 items

b. (1) $H_0: \mu_1 = \mu_2 = \mu_3 = \mu_4$

H_A: at least 1 treatment mean is different

(2) $\alpha = 0.05$

(3) $F = \frac{MST}{MSE} \sim F_{(k-1),(b-1)(k-1);0.05}$

3 21

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(4) $F^* = 26.52 > F_{3,21;0.05} = 3.07$

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(5) R.R: $\{F \mid F > F_{3,21;0.05}\}$
(6) ∴ Reject H_0

c. (1) $H_0: \mu_1 = \mu_2 = \dots = \mu_8$
$H_a: \exists\, \mu_i \neq \mu_j \ (i \neq j)$
(2) $\alpha = 0.05$
(3) $F = \frac{MSB}{MSE} \sim F_{(b-1),(k-1)(b-1);0.05}$
(4) $F^* = 35.32 > F_{7,21;0.05} = 2.4876$
(5) R.R: $\{F \mid F > F_{7,21,0.05}\}$
(6) ∴ Reject H_0

C11S04Q25
25. a. $q_{0.05}(4,21) = 3.95$
k, df = n-k

b. $w = q_{0.05}(4,21) \times \sqrt{\frac{0.1658}{8}} = 0.569$

c. $|\bar{x}_1 - \bar{x}_2| = 0.4875 < 0.569$ => no differ
$|\bar{x}_1 - \bar{x}_3| = 0.205 < 0.569$ => no differ
$|\bar{x}_1 - \bar{x}_4| = 1.658 > 0.569$ => differ
$|\bar{x}_2 - \bar{x}_3| = 0.28 < 0.569$ => no differ
$|\bar{x}_2 - \bar{x}_4| = 1.171 > 0.569$ => differ
$|\bar{x}_3 - \bar{x}_4| = 1.45 > 0.569$ => differ

C11S04Q26
26. a. Treatment = mixture, block = investigator

Source	SS	df	MS	F
Treatment	107.2	2	53.6	12.32
Block	176.4	4	44.1	10.14
Error	34.8	8	4.35	
Total	318.4	14		

b. (1) $H_0: \mu_1 = \mu_2 = \mu_3$
$H_a: \exists\, \mu_i \neq \mu_j \ (i \neq j)$
(2) $\alpha = 0.01$

No.:

③ $F = \frac{MST}{MSE} \sim F_{\overset{2}{(k-1)},\overset{8}{(k-1)(b-1)};0.01}$

④ $F^* = 12.32 > F_{2,8;0.01} = 8.65$

⑤ $R.R: \{F \mid F > F_{2,8;0.01} = 8.65\}$

⑥ $\therefore$ Reject H_0

c. $P(F^* > F_{4,8;0.01}) = P(10.14 > 7.01)$

Because $10.14 > 7.01$, the p-value for blocks is less than 0.01

$\therefore$ Yes, Blocking was highly effective. There was a real difference in the 5 investigators.

d. $w = q_{0.01}(3,8) \times \sqrt{\frac{4.35}{5}} = 5.06$

$|\bar{T}_1 - \bar{T}_2| = 2 < 5.06 \Rightarrow$ no differ

$|\bar{T}_1 - \bar{T}_3| = 6.4 > 5.06 \Rightarrow$ differ

$|\bar{T}_2 - \bar{T}_3| = 4.4 < 5.06 \Rightarrow$ no differ

C11503014

14. • Method 2-1: • (-0.1438, -0.0242) => doesn't contain zero
• P-value = 0.007 < 0.05
• There is a significance difference

• Method 3-1: • (-0.0178, 0.1018) => contains zero
• P-value = 0.129 > 0.05
• There is no significance difference

• Method 3-2: • (0.0662, 0.1858) => doesn't contain zero
• P-value = 0 < 0.05
• There is a significance difference

∴ Method 2 & 1, 3 & 2 have a significant difference, while method 3 & 1 doesn't have a significant difference.